The Effect of Bye Weeks on Injury Event Rates in the Canadian Football League

Caitlin Lee, MD,* Brice Batomen, PhD,† Dhiren Naidu, MD,* Shane Hoeber, MD,* Robert McCormack, MD,‡ Russell Steele, PhD,§ Arijit Nandi, PhD,§ and Ian Shrier, MD, PhD§

Abstract

Objective: To determine the effect of bye weeks (no practices or games) on the injury event rate in the Canadian Football League (CFL). **Design:** Historical (retrospective) cohort study. **Setting:** CFL. **Participants:** CFL athletes between 2011 and 2018. **Intervention:** CFL pseudorandom assignment of bye weeks each season (2011-2013: 1; 2014-2017: 2; 2018: 3). **Main Outcome Measures:** Game injury incident rate ratio (IRR) in the week following a bye week compared with non–bye weeks. Sensitivity analyses: IRR for the 2 and 3 weeks following a bye week. We conducted exploratory analyses for combined game and practice injury events because we did not have the number of players exposed during practice. **Results:** The IRR was 0.96 (0.87-1.05), suggesting no meaningful effect of a bye week on the post–bye week game injury event rate. We obtained similar results for cumulative game injury events for subsequent weeks: IRR was 1.02 (0.95-1.10) for the 2 weeks following the bye week and 1.00 (0.93-1.06) for the 3 weeks following the bye week. The results were similar with 1, 2, or 3 bye weeks. However, the combined game and practice injury event rate was increased following the bye week [IRR = 1.14 (1.05-1.23)]. These results are expected if the break period results in medical clearance for preexisting injuries; increasing pain in these locations following the bye week would now be considered new injuries instead of "exacerbations." **Conclusions:** Bye weeks do not appear to meaningfully reduce the injury event rate. Furthermore, there was no injury reduction when adding additional bye weeks to the schedule.

Key Words: Canadian Football League, football, injury rate, bye week

(*Clin J Sport Med* 2024;00:1–6)

INTRODUCTION

Injuries in Canadian and American football are common and represent a significant area of interest for many players, coaches, clinicians, and even spectators of the sport. In a study that compared 15 National Collegiate Athletic Association (NCAA) sports, football had the highest injury event rates overall.[1] New and ongoing injuries that occur throughout a season provide a substantial burden on teams and can affect the performance of a team as a whole.[2] A few other studies on NCAA football demonstrated that injury event rates change throughout a season.[3–5] In 2017, Robbins et al described injury event rates and injury patterns for the first time in the Canadian Football League (CFL). Rates of new injuries in the CFL declined by 1% per week in the regular season, with a substantial 30% decline in reported injury event rates during the playoffs. Although injury event rates decreased throughout the season, the number of players that were injured in any given week (injury burden) increased throughout the regular season.[2] This information is highly valuable in aiding the decisions of team staff and physicians throughout the season.

Coaches and clinicians are continuously trying to find ways to mitigate the risk of injury for their players. An abundance of research exists on different methods of injury prevention for numerous sports, including football.[6–8] Rest periods are thought to be an important intervention that could reduce an athlete's risk of injury[9,10] and may also provide relief of physical and mental fatigue.[9,11,12]

The CFL defines a bye week as generally having no games for at least 10 days and no training sessions or practices for at least 7 days. Although bye weeks in the CFL were necessary when there were an odd number of teams in the league, they also presented an opportunity for mandated rest periods for the players. Bye weeks were negotiated by the CFL Players' Association (CFLPA) as a break to avoid short weeks where a team might have to play twice in 1 week. Some authors have postulated that bye weeks may reduce injury risk,[13,14] and it is also possible that they increase injury event rates because teams play more aggressively before or after the bye week. Teramoto et al[15] looked at 582 concussions in the National Football League (NFL) and found no significant association between the number of concussions that occurred and the

Submitted for publication December 25, 2023; accepted August 9, 2024.

*From the *University of Alberta, Edmonton, Canada; †University of Toronto, Toronto, Canada; ‡University of British Columbia, Vancouver, Canada; and §McGill University, Montreal, Canada.*

Dr. D. Naidu is a Chief Medical Officer (CMO) for the CFL and a team physician for the Edmonton Elks. Dr. R. McCormack is a CMO for the CFL, a team physician for the BC Lions, and a member of the Editorial Board for CJSM. Dr. I. Shrier and Dr. R. Steele are recipients of Canadian Institutes of Health Research grant funding to improve research and statistical methods for studying sport injuries. The funding is unrelated to this manuscript. The remaining authors do not have any disclosures related to the submitted work.

We will make summary data available but do not have the authority to release raw data because it is owned by a private company and it might allow someone to identify individual participants.

Corresponding Author: Caitlin Lee, MD, 3514 Allan Drive SW #201, Edmonton, AB T6W 3G9 (cmlee@ualberta.ca).

Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (www.cjsportmed.com).

Copyright © 2024 Wolters Kluwer Health, Inc. All rights reserved.

http://dx.doi.org/10.1097/JSM.0000000000001272

Copyright © 2024 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

number of rest days or timing of the bye week by the team or the opponent. However, their analysis compared concussion rates post–bye week to an average concussion rate of the entire season, without accounting for varying injury event rates over the season. The CFL has added bye weeks to their regular seasons between 2011 and 2019 for various scheduling purposes. However, the relationship between bye weeks and injury event rates is still unknown. This is of importance as certain game schedules may lead to an increased risk of injuries.

The primary objectives of our study were to examine the effect of bye weeks on game injury event rates and to determine how many weeks such an effect might last. Secondary objectives included examining (1) whether the number of bye weeks or timing of the bye week in a season modifies the injury event rate and (2) estimating combined game and practice injury event rates.

METHODS

We used a historical (retrospective) cohort design. We obtained data on player participation in games through the CFL injury surveillance database that included active and inactive players on each team roster for each regular season game.[16]

All active CFL players in the 2011 to 2018 seasons were included. There were 8 teams for the 2011 to 2013 seasons and 9 for the 2014 to 2018 seasons. Only regular season games were considered; preseason and playoff games were excluded because the style of play is likely fundamentally different from regular season games, and they do not include "bye weeks" even if the team is not playing. We did not analyze bye weeks that occurred in the last week of the regular season because there were no post–bye week games for comparison. Data indicating which players attended practices were unavailable; it was assumed all players participated in all practices, and all teams had the same number of players and practices.

Intervention

The CFL defines a bye week as generally having no games for at least 10 days and no training sessions or practices for at least 7 days. Within a bye week, there are no organized training sessions, practices, or games. Teams were allotted bye weeks in their schedules through pseudorandom assignment each season by the CFL. In brief, the allocation was based on stadium availability. In years with more than 1 bye week, attempts were made to distribute the bye weeks evenly throughout the season if possible. Performances or injuries in past years were not included in the decision-making process.

The regular season game schedules from 2011 to 2018 were obtained from the official CFL website. See **Supplemental Digital Content 1-3 (see Figures**, http://links.lww.com/JSM/A451, http://links.lww.com/JSM/A452, http://links.lww.com/JSM/A453) for the timing of bye weeks in each of the 8 seasons. The 2011 to 2013 CFL seasons contained 1 bye week per team, the 2014 to 2017 seasons generally had 2 bye weeks per team (with 1 team having 3 bye weeks in each year) and the 2018 season had 3 bye weeks per team.

Outcome

Injuries

The CFL injury surveillance database includes electronic health records (ARMS, https://edge10group.com/product/health/) based on prospectively collected data entered directly by team physicians/therapists and recorded on a daily basis for all players who reported to medical staff for assessment and/or treatment. Injury information included: player position, injury date, recovery date, diagnosis, anatomical location, and game versus practice injury events. We excluded preventive medical appointments and non-football, weight room, and off-season injuries.

An injury event was considered if a player presented for assessment/treatment of an injury and a treatment note was made by medical staff.[2] We included all musculoskeletal and neurological game-related and practice-related injuries in the regular season. If a team practiced the day(s) following a game, the practice injury event was considered to have occurred in the week following the game, even if the practice was in the same "game calendar week" according to the CFL schedule. We excluded illnesses, and other nonmusculoskeletal injuries including cardiovascular, respiratory, dermatology, endocrine, endocrine, environmental, fluid & electrolyte, gastrointestinal, genitourinary, hematology, infectious disease, psychology, and those that did not include a description or categorization.

Injury Event Rate

Our primary analysis was restricted to game-related injury event rates. If an athlete incurred 2 injuries during the same incident, we considered this a single injury event. We used the athletes at risk (AAR) approach (expressed as injuries per 100 AAR) to account for the number of AAR because it most closely estimates the true injury event rate.[17] In the CFL, there are 12 AAR for each team at any point in time; using the AAR approach, the total number of AAR in 1 game is 12 athletes per team. The weekly injury event rate is then equal to the number of injury events/AAR.

Statistical Analysis

For our primary analysis, we estimated incidence rate ratios (IRRs, 95% CI) for game injury events in the week following a bye week compared with non–bye weeks with conditional quasi-Poisson regressions to account for overdispersion of events.[18] Based on our directed acyclic graph (see **Figure, Supplemental Digital Content 4**, http://links.lww.com/JSM/A454), estimating the effect of bye weeks might not require any adjustment for confounders because the bye weeks are essentially randomly assigned. However, it has been established that injury event rates vary based on the time of the season. We accounted for known trends in injury event rates over the season and other factors by adjusting for (1) the timing of the game within a season (game number), (2) the opposing team, and (3) the game location (home vs away). We also included a variable "Available Practice Days," to account for the fact that the interval between games varies between 4 and 13 days when bye weeks are not present. In addition, teams may change playing style, coaches, and players each

Copyright © 2024 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

year. To account for these factors, we used a single combined variable "team-year" that identified each team in each year as "fixed-effects," with their specific intercepts treated as nuisance parameters. To account for the autocorrelation in injury events rates between games, we added lagged deviance residuals in the model.[18]

To explore how long the effect of a bye week might last, we computed the cumulative injury event rate for 2 and 3 weeks following a bye week in addition to our primary analysis using only the week after the bye week.

We had 2 secondary objectives. First, we explored whether the number of bye weeks in a season modified our effect of interest by (1) presenting separate results for seasons that had 1, 2, and 3 bye weeks, and (2) restricting the data to the 2014 to 2017 seasons where each team had 2 bye weeks, and assessed the effect of the first and second bye week separately by using only pre–bye weeks as control. Second, we repeated our primary analysis with both game and practice injury event rates.

Sensitivity Analysis

Our approach assumes that game weeks during a regular season within a given team were exchangeable each year after we accounted for variables and time trends known to impact injury event rates (listed above). To further assess the robustness of our results, we conducted analyses limiting control weeks to the 3 weeks preceding a bye week.

Exploratory Analyses for Potential Mechanisms

To determine whether there might be higher injury event rates in the week before the bye week because teams knew they were going to have an extended rest we compared the injury event rate in the week before the bye week, the second week before the bye week, and the third week before the bye week.

All analyses were conducted in R for Statistical Computing, using the R packages X, Y, and Z. The R code is available on OSF (https://osf.io/p2htd/?view_only=271e382a4ce041209b72e0dcd9ffc261).

Ethical Considerations

This study used prospective data collected as part of normal CFL business practice. According to Article 2.5 of the Canadian Tri-Council Policy Statement on the Ethical Conduct for Research Involving Humans,[19] this represents a quality assurance project and does not require ethics approval.

RESULTS

Original data from 8 CFL seasons comprised 7642 game injury events. We excluded 213 duplicate game injury events, 310 injuries with missing data on which team the athlete played for or whether it was game/practice related and 882 game injury events that occurred during preseason and playoffs. There were 6236 game injury events available for our primary analysis (Figure 1). The type of game injuries and location of game injuries that occurred are illustrated in Table 1.

Our primary analysis suggested that a bye week was associated with a 4% reduction [IRR = 0.96 (95% CI, 0.87-1.05)] in the game injury event rate in the following week, although this was estimated with some uncertainty (see **Table, Supplemental Digital Content 1**, http://links.lww.com/JSM/A455, top). Our results also suggest, with similar uncertainty, that the apparent reduction only lasts 1 week, as the IRR was 1.02 (95% CI, 0.95-1.10) and 1.00 (95% CI, 0.93-1.06) for the cumulative injury event rate 2 weeks (see **Table, Supplemental Digital Content 1**, http://links.lww.com/JSM/A455, middle) and 3 weeks (see **Table, Supplemental Digital Content 1**, http://links.lww.com/JSM/A455, bottom) following the bye week, respectively.

We had 2 secondary objectives. First, we assessed the potential effect of the number of bye weeks by season (Figure 2A, (see **Table, Supplemental Digital Content 1**, http://links.lww.com/JSM/A455). Although our results might suggest there was a decreasing protective effect for injury events with an increasing number of bye weeks, there was only 1 year of data for 3 bye weeks, and the confidence intervals were large for all comparisons, reflecting uncertainty around the estimates. Our sensitivity analysis to account for potentially inadequate control of known time trends suggested bye weeks were not protective against injury events (Figure 2B, (see **Table, Supplemental Digital Content 4**, http://links.lww.com/JSM/A458).

Finally, when we restricted the data to years including 2 bye weeks (2014-2017) and only used pre–bye weeks as control, the IRR for the first bye week which occurred early in the season [1.05 (95% CI, 0.84-1.31)] was not distinguishable from the IRR for the second bye week which occurred later in the season [1.04 (95% CI, 0.84-1.29)] (see **Table, Supplemental Digital Content 2**, http://links.lww.com/JSM/A456).

When we included both game and practice injury events, there was an increased injury event rate in the week following the bye week [IRR = 1.14 (95% CI, 1.05-1.23)] (see **Table, Supplemental Digital Content 3**, http://links.lww.com/JSM/A457). The increased injury event rate was also observed when we considered the cumulative injury event rate in the 2 weeks [IRR = 1.13 (95% CI, 1.07-1.20)] and 3 weeks [IRR = 1.08 (95% CI, 1.02-1.14)] following a bye week.

Exploratory Mechanistic Analyses

Unexpectedly, the injury event rate during the week immediately before the bye week was generally lower than the preceding weeks. The injury event rate was 5.2 (95%CI: 3.7-7.2) in the third week before the bye week, 5.0 (95% CI, 3.6-7.0) in the second week before the bye week, and 4.2 (95% CI, 3.0-5.9) in the week before the bye week.

DISCUSSION

Overall, there was no evidence that the inclusion of bye weeks meaningfully reduces the overall injury event rate. Although our initial analysis suggested that bye weeks might reduce injury event rates, the effect was negated when we restricted "control" periods to the 3 game weeks preceding the bye week, and there was no indication of cumulative effects. Finally, subgroup analyses indicate that the timing and the number of bye weeks do not modify game injury events.

Our findings are similar but more nuanced than a recent NFL study where bye weeks did not increase the rate of concussions.[15] Another professional football study by Perez et al[20] found that shorter rest periods (<4 days) associated

Copyright © 2024 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

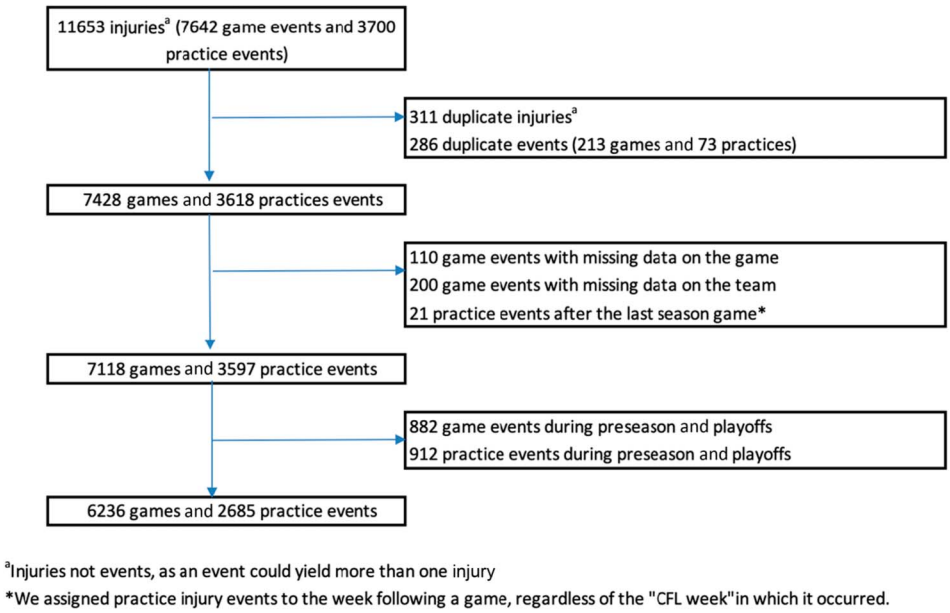

Figure 1. Flow diagram illustrating number of documented injury events and exclusions due to duplication, missing data, and exclusion criteria.

with Thursday night games in the NFL do not increase rates of game injuries.

Exploratory analyses using both game and practice events suggest an increase in practice injury event rates following a bye week. This is expected if the bye week leads to increased medical clearance for already existing injuries because any increase in symptoms would now be considered a new injury instead of an exacerbation. Injured players are also more likely to return to a practice setting before a full game, which could explain the discrepancy in the data once practices were included. Furthermore, there may also be a deconditioning effect that occurs during a bye week making players more prone to injury when returning to practice the following week. Given the results of this study and the potential contribution of a deconditioning effect, coaches could consider reducing the intensity/duration of initial post–bye week practices to counteract any possible effect.

TABLE 1. Game Injury Characteristics

Type/Location	Category	% (n)
Injury type	Joint/Ligament	32.0 (1993)
	Muscle/Tendon	22.9 (1426)
	Contusions	23.1 (1443)
	CNS/PNS	7.5 (467)
	Bone (208 fractures, 20 others*)	3.6 (228)
	Other	10.9 (679)
Anatomical locations	Lower leg, ankle, and foot	22.2 (1384)
	Hip, groin, and thigh	16.9 (1052)
	Upper extremity	16.0 (997)
	Knee	14.7 (915)
	Head and neck	11.4 (713)
	Shoulder	11.1 (693)
	Trunk	7.6 (476)
	Nonspecific/missing	0.1 (6)

** Other game-related bone injuries included accessory navicular, sesamoiditis, heel spur, periosteal reactions, spinal stenosis, and spondylolisthesis.*

Although injury event rates increased following a bye week once practice data were added, it is worth noting that the total number of game-related injuries (6236) was still much higher than practice-related injuries (2685). Given each team has approximately 1 game per week and 3 to 5 practices, our data show the game injury risk is still much higher than the practice injury risk. This is similar to previous studies on NCAA football.[3–5]

Despite our study showing injury event rates were unaffected by bye weeks, other potential benefits should be considered. There may be mental health benefits for players having a designated time to spend with their family, friends, and teammates. Furthermore, the extra time allows teams to focus on game preparation strategies (eg, studying their play books or coming up with new plays), which may explain why NFL teams have an increased winning percentage following longer rest periods.[21]

Our work has several limitations. First, the data we used are from the CFL injury data surveillance program, which is not primarily designed for research. However, this information was part of the CFL-mandated medical documentation of therapists and physicians, is regularly audited, and is likely to be as complete as possible. Second, we used player roster lists for each game but some players listed may not have actually played in the game. However, we used the AAR method (described above) that avoids underestimating true risk simply because there are "active" players who do not play. Third, any violation of the key assumptions underlying our statistical models would likely bias our results. Our first key assumption is that the exposure of interest (bye week) is well defined. Because bye weeks are "pseudorandomly assigned," this is likely satisfied. Our second key assumption is that non–bye weeks and bye weeks were exchangeable. We cannot rule out the possibility that game weeks were not

Copyright © 2024 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

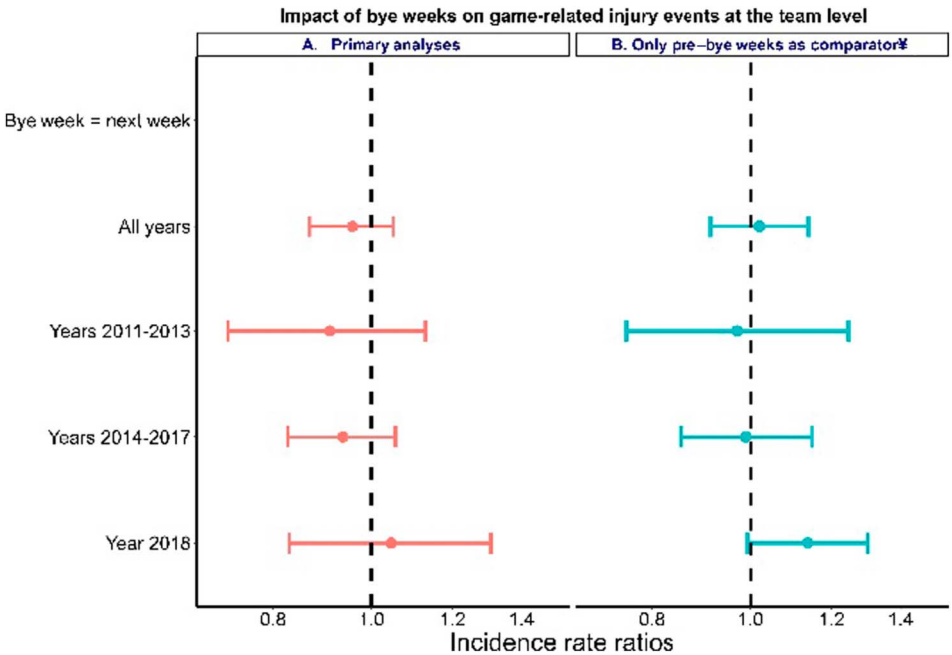

Figure 2. Incidence rate ratio (IRR) comparing game-related injury events during weeks following a bye week to non–bye weeks. Panel A results are a conditional Poisson model adjusted for the secular trends in injury event rates over a season, the opposing team, an indicator to identify whether a team was playing at home or away, and the "available practice days" between games. "Team-years" was used as the stratum to account for differences between among teams during the same year and between the same team over the years. The top section compares the injury event rate in the week before the bye week with the week after the bye week for all years and is stratified according to how many bye weeks occurred each year. Panel B illustrates a similar analysis to control for secular trends except we restricted the control weeks to game weeks preceding a bye week, with a maximum of 3 controls per bye week.

exchangeable, despite our approach accounting for measured confounding factors and time trends. In fact, our sensitivity analyses yielded slightly different results from our primary analysis, suggesting residual time-varying confounding remained in our primary analysis. This fits with previous findings that injury event rates decline over the course of a season in the CFL.[2] One reason for this trend is that players who are more prone to injury are injured sooner.[22] Therefore, the overall risk of the remaining players will be lower. In our analyses, the injury event rate decreased over the 3 weeks preceding a bye week. Finally, we could not evaluate whether the bye week affected strategic coaching and injury reporting decisions.

Further research on the effects of the bye week in football should attempt to minimize charting inconsistencies of new injuries versus recurrent or reaggravated injuries, as this distinction may impact results.[23] Other areas of interest that were beyond the scope of this study and would require a much larger sample size might include subgroup analyses between (1) traumatic versus nontraumatic injuries, (2) different positions (eg skill-based vs non-skill-based players), (3) presence/absence of previous injury in the current season, and (4) the effect of bye weeks on illnesses such as heat-related illness and mental health. Further work needs to be completed in this area before making definitive conclusions regarding the overall impact of bye weeks. Given the many ways bye weeks can impact the game, players, and their health, these results should be interpreted as part of a larger picture and not be used in isolation to determine the timing or number of bye weeks instituted by leagues.

CONCLUSIONS

The injury event rate in the CFL was not meaningfully different following a bye week compared with non–bye weeks. The total number of bye weeks within a season does not appear to impact team injury event rates. Overall, these findings provide coaches/players with a better understanding of injury risk surrounding bye weeks within a season. Although bye weeks do not appear to impact the injury event rate, they may have other potential noninjury-related benefits (eg mental health, performance).

References

1. Hootman JM, Dick R, Agel J. Epidemiology of collegiate injuries for 15 sports: summary and recommendations for injury prevention initiatives. *J Athl Train*. 2007;42:311–319.
2. Robbins SM, Bodnar C, Donatien P, et al. The influence of time of season on injury event rates and the epidemiology of Canadian football injuries. *Clin J Sport Med*. 2020;1.
3. Krill MK, Borchers JR, Hoffman JT, et al. Effect of exposure type and timing of injuries in division I college football: a 4-year single program analysis. *Phys Sportsmed*. 2017;45:26–30.
4. Dick R, Ferrara MS, Agel J, et al. Descriptive epidemiology of collegiate men's football injuries: national collegiate athletic association injury surveillance system, 1988-1989 through 2003-2004. *J Athl Train*. 2007; 42:221–233.
5. Kerr ZY, Simon JE, Grooms DR, et al. Epidemiology of football injuries in the national collegiate athletic association, 2004-2005 to 2008-2009. *Orthop J Sports Med*. 2016;4:2325967116664500.
6. Emery CA, Pasanen K. Current trends in sport injury prevention. *Best Pract Res Clin Rheumatol*. 2019;33:3–15.
7. Kirkendall DT, Junge A, Dvorak J. Prevention of football injuries. *Asian J Sports Med*. 2010;1:81–92.
8. Schiff MA, Caine DJ, O'Halloran R. Injury prevention in sports. *Am J Lifestyle Med*. 2010;4:42–64.
9. Orlando C, Levitan EB, Mittleman MA, et al. The effect of rest days on injury rates. *Scand J Med Sci Sports*. 2011;21:e64–e71.
10. Johnston CA, Taunton JE, Lloyd-Smith DR, et al. Preventing running injuries. Practical approach for family doctors. *Can Fam Physician*. 2003; 49:1101–1109.
11. Kentta G, Hassmen P. Overtraining and recovery. A conceptual model. *Sports Med*. 1998;26:1–16.
12. Cosca DD, Navazio F. Common problems in endurance athletes. *Am Fam Physician*. 2007;76:237–244.
13. Howington EB, Moates KN. Is there a bye week advantage in college football?. *Electron J Appl Stat Anal*. 2017;10:735–744.

Copyright © 2024 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

14. Sung YT, Tainsky S. The National football league wagering market: simple strategies and bye week–related inefficiencies. *J Sports Econ*. 2014; 15:365–384.
15. Teramoto M, Cushman DM, Cross CL, et al. Game schedules and rate of concussions in the national football league. *Orthopaedic J Sports Med*. 2017;5:2325967117740862.
16. Gasparrini A. The case time series design. *Epidemiology (Cambridge, MA)*. 2021;32:829–837.
17. Stovitz SD, Shrier I. Injury event rates in team sport events: tackling challenges in assessing exposure time. In: *BMJ Publishing Group Ltd and British Association of Sport and Exercise Medicine*; 2012:960–963.
18. Armstrong BG, Gasparrini A, Tobias A. Conditional Poisson models: a flexible alternative to conditional logistic case cross-over analysis. *BMC Med Res Methodol*. 2014;14:122.
19. Canada Go. *Tri-council Policy Statement: Ethical Conduct for Research Involving Humans–TCPS 2*. Ottawa, Canada: Canadian Institute of Health Research; 2018.
20. Perez JR, Burke J, Zalikha AK, et al. The effect of Thursday night games on in-game injury rates in the national football league. *Am J Sports Med*. 2020;48:1999–2003.
21. Foreman J, Soebbing B, Rodenberg R. The effect of time-off and familiarity on organizational performance: evidence from the National Football League. *Int J Sport Manag*. 2016;17:84–101.
22. Shrier I, Zhao M, Piché A, et al. A higher sport-related reinjury risk does not mean inadequate rehabilitation: the methodological challenge of choosing the correct comparison group. *Br J Sports Med*. 2017;51:630–635.
23. Hamilton G, Meeuwisse WH, Emery C, et al. Subsequent injury definition, classification, and consequence. *Clin J Sport Med*. 2011;21:508–514.

Copyright © 2024 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.